

Foundation (Jalapeno)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{2}}$.

- (A) $\frac{\sqrt{2}}{2}$
- (B) $\frac{1}{2\sqrt{2}}$
- (C) $\frac{2}{\sqrt{2}}$
- (D) $\frac{\sqrt{2}}{\sqrt{2}}$
- (E) $\frac{1}{2}$

Q2. Which is the rationalized form of $\frac{1}{\sqrt{3}}$?

- (A) $\frac{1}{3}$
- (B) $\frac{\sqrt{3}}{3}$
- (C) $\frac{\sqrt{3}}{6}$
- (D) $\frac{1}{\sqrt{3}}$
- (E) $\frac{3}{\sqrt{3}}$

Q3. Rationalize the denominator of $\frac{2}{\sqrt{5}}$.

- (A) $\frac{2\sqrt{5}}{5}$
- (B) $\frac{\sqrt{5}}{5}$
- (C) $\frac{2}{5}$
- (D) $\frac{2}{5\sqrt{5}}$
- (E) $\frac{5}{2\sqrt{5}}$

Q4. What is the rationalized form of $\frac{3}{\sqrt{2}}$?

- (A) $\frac{3}{2}$
- (B) $\frac{3\sqrt{2}}{2}$
- (C) $\frac{2}{3}$
- (D) $\frac{\sqrt{2}}{3}$
- (E) $\frac{3}{2\sqrt{2}}$

Q5. Rationalize the denominator of $\frac{\sqrt{7}}{\sqrt{3}}$.

- (A) $\frac{\sqrt{21}}{3}$
- (B) $\frac{\sqrt{21}}{7}$
- (C) $\frac{7}{\sqrt{3}}$
- (D) $\frac{3}{\sqrt{7}}$
- (E) $\frac{\sqrt{3}}{7}$

Q6. Which is the rationalized form of $\frac{1}{\sqrt{6}}$?

- (A) $\frac{\sqrt{6}}{6}$
- (B) $\frac{\sqrt{6}}{3}$
- (C) $\frac{1}{6}$
- (D) $\frac{6}{\sqrt{6}}$
- (E) $\frac{\sqrt{6}}{2}$

Q7. Rationalize the denominator of $\frac{2\sqrt{3}}{\sqrt{2}}$.

- (A) $\frac{2\sqrt{6}}{2}$
- (B) $\frac{2\sqrt{3}}{2}$
- (C) $\frac{\sqrt{6}}{2}$
- (D) $\frac{2\sqrt{2}}{\sqrt{3}}$
- (E) $\frac{2\sqrt{3}}{\sqrt{2}}$

Q8. What is the rationalized form of $\frac{\sqrt{5}}{\sqrt{10}}$?

- (A) $\frac{\sqrt{2}}{2}$
- (B) $\frac{\sqrt{10}}{10}$
- (C) $\frac{\sqrt{50}}{10}$
- (D) $\frac{\sqrt{2}}{10}$
- (E) $\frac{\sqrt{2}}{5}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{2+\sqrt{3}}{\sqrt{2}-1}$. Show your work and simplify your answer.

Q10. Rationalize the denominator of $\frac{1}{\sqrt{7}-\sqrt{3}}$. Show your work and simplify your answer.

Chef's Tips

- Emphasize the importance of multiplying by the conjugate to rationalize the denominator.
- Make sure students simplify their answers.
- Remind students that when rationalizing the denominator, they should not leave any radicals in the denominator.